

Vision–Language Models Read Their Own Attention at the Wrong Depth

Anonymous authors
Paper under double-blind review

Abstract

Vision–language models decide what to look at — which visual tokens to keep, where to crop — by reading their own attention. We show the field reads it at the wrong depth. Question-conditioned localisation does not exist in the vision encoder (at chance on four architectures), is too weak at early language-model layers to be usable, and becomes reliable only in the upper half of the stack. The practical cost is large and, to our knowledge, unreported: ranking visual tokens for pruning by layer-2 attention — the default in a widely used method — is **worse than ranking them at random** on both models we test, losing 22.0 pp and 13.1 pp against no pruning at all. Reading late layers instead discards **90% of visual tokens at no measurable cost** (−0.5 pp and −1.6 pp), a margin of +24.1 pp and +11.5 pp over the early-layer default, with no training. The same defect appears in crop placement: replacing the standard block-mean read-out with a learned one improves evidence localisation on three of four architectures spanning two model families, and question answering by +12.0 pp and +10.5 pp over an unmodified model. A logit-lens analysis shows why cropping helps at all: it does not improve reasoning but restores an answer-formation step that appears abruptly at a single layer, and only when the crop actually delivers the evidence. We report what does not work with equal weight, including seven internal interventions that all fail and five of our own claims rejected by cross-architecture replication.

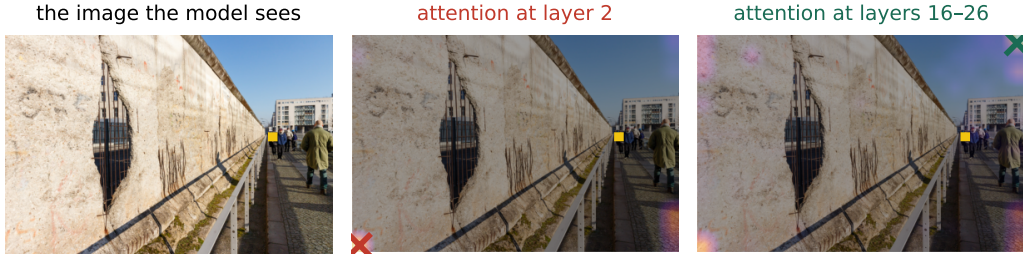
1 Introduction

A vision–language model (VLM) turns an image into a few hundred visual tokens. When the question concerns something small, the evidence occupies a fraction of one token and the model cannot answer: below roughly 0.15 merged tokens of target extent we measure 19.0% accuracy — below the 25% chance level of a four-way question — while the identical question asked of the resolved region is answered correctly 96–100% of the time.

Two large families of methods respond to this, and both make a decision from the model’s attention. Token pruning discards visual tokens judged unimportant. Crop-and-re-encode methods select a region and look again. Both read attention, and the depth at which they read it has been treated as an implementation detail. It is not.

Our contributions:

- Pruning at the wrong depth is worse than not ranking at all. Ranking visual tokens by layer-2 attention scores below random selection on two architectures. A late read-out at the same budget retains the accuracy of using every token while discarding 90% of them (Section 5).
- The standard read-out averages away its own signal. Collapsing a block of layers into one map and taking the maximum is worse than a learned combination on three of four architectures, two model families (Section 4).
- A mechanism, with a causal control. Cropping does not improve reasoning: cropped and uncropped trajectories are indistinguishable for twenty layers and then separate



"What is the color of the woman's scarf?"

Figure 1: The same model, the same forward pass, read at two depths. A real V*Bench item with the queried object boxed in yellow. Middle and right show the model’s actual attention over image positions, upsampled onto the image it received; × is each read-out’s argmax. At layer 2 — where the standard pruning default reads — the mass sits away from the target. At layers 16–26 it concentrates on it. Nothing here is illustrative: both maps come from one inference pass.

abruptly — and the separation appears only on items where the crop actually contains the evidence (Section 6).

- An audited account of failure. Seven internal interventions that would have been free if they worked, all null; and five claims of our own rejected when tested on a second architecture (Section 7, Appendix B).

2 Setup

All experiments use V*Bench ($n=191$ four-way questions with ground-truth boxes) and, for transfer, HR-Bench 4k ($n=800$, CircularEval, 4032² images). Models: Qwen3-VL-2B, Qwen2-VL-7B, LLaVA-OneVision-7B, LLaVA-NeXT-7B. Token counts are always measured from the processor’s realised grid, never computed; any arm drifting more than 10% from its target voids its contrast. Every learned component is fitted out-of-fold with folds grouped by item. Appendix A gives the full protocol.

3 Related work

Token pruning. FastV (Chen et al., 2024) discards visual tokens judged unimportant by attention received at an early layer; its default reads layer 2. Our Section 5 is a direct measurement of what that depth choice costs.

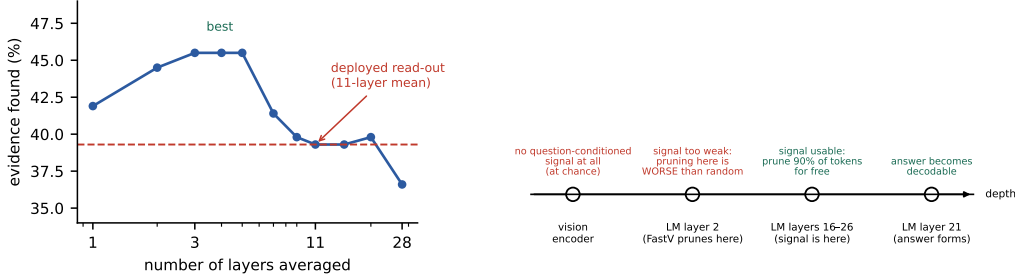
Crop-and-re-encode. V* (Wu & Xie, 2024) and its successors select a region from model attention and re-encode it. These methods are scored here against what the same realised token budget buys as one uniform image, a comparison the literature does not report (Appendix G).

Layer-contrastive decoding. DoLa (Chuang et al., 2024) and visual contrastive decoding (Leng et al., 2024) exploit differences between layers or inputs at decoding time. We test both families and find them null here (Appendix K); the reason, we argue, is that no decoding-time operation can supply information the encoder never wrote.

Attention as explanation. Jain & Wallace (2019) argue attention weights are not explanations. Our results are compatible but more specific: attention read at the right depth carries a strong, causally-verifiable localisation signal, while the same attention read at an early layer is worse than random. Attention sinks (Xiao et al., 2024) are a known phenomenon in

Table 1: Evidence localisation, four architectures, two families. “Learned” is a ridge reweighting of per-layer maps, fitted out-of-fold. Chance is $\approx 2.2\%$ throughout.

MODEL	BLOCK MEAN	LEARNED	Δ	95% CI
Qwen3-VL-2B	39.3%	45.5%	+6.3 pp	[+2.6, +10.5]
Qwen2-VL-7B	35.1%	43.5%	+8.4 pp	[+3.1, +13.6]
LLaVA-OneVision-7B	25.1%	26.2%	+1.0 pp	[−2.6, +5.2]
LLaVA-NeXT-7B	12.6%	18.8%	+6.3 pp	[+2.1, +10.5]



(a) Averaging the k individually-best layers. More is worse past $k \approx 3$, and the deployed 11-layer mean sits well down the curve. (b) Where the question-conditioned signal lives, and what it costs to read it too early.

Figure 2: The defect (left) and its location in the network (right).

language models; Appendix E documents a visual analogue on the serialisation boundary across four architectures, and shows it is not the cause of the read-out failure.

4 The read-out defect

The standard way to locate evidence in a VLM is to average attention over a block of layers and take the argmax. Figure 1 shows why this is lossy: layers disagree about where the target is, and a mean cannot represent disagreement.

Three of four clear zero; LLaVA-OneVision is a null and we report it as one. It is also the only model whose best single layer equals its block mean, i.e. whose layers agree — internally consistent, but a single data point we do not build on. Absolute localisation varies by a factor of three across models, which matters for anyone porting crop methods to LLaVA.

5 Pruning at the wrong depth

Token pruning ranks visual tokens by attention and drops the low-ranked ones. The widely used default reads that attention at layer 2. Figure 3 shows the consequence. Every arm prunes the same number of tokens, so cost is matched by construction.

At 10% keep, the late read-out is within noise of using every token (−0.5 pp [−5.8, +4.7] and −1.6 pp [−6.3, +3.1]), while layer-2 ranking loses 22.0 pp [−29.8, −14.7] and 13.1 pp respectively and falls below random selection on both models (−3.7 pp, −4.7 pp). The margin is +24.1 pp [+16.2, +31.9] and +11.5 pp [+4.2, +18.8].

No learning is required. A plain average of late layers matches a trained signed combination at every operating point (+2.6, −1.0, +1.0 pp, all null). The recommendation is a one-line change of which layer index is read.

A prediction that failed. We pre-registered that the penalty would be larger on Qwen2-VL, whose early layers rank the target worse. It is smaller (−13.1 vs −21.9 pp). Layer ranking

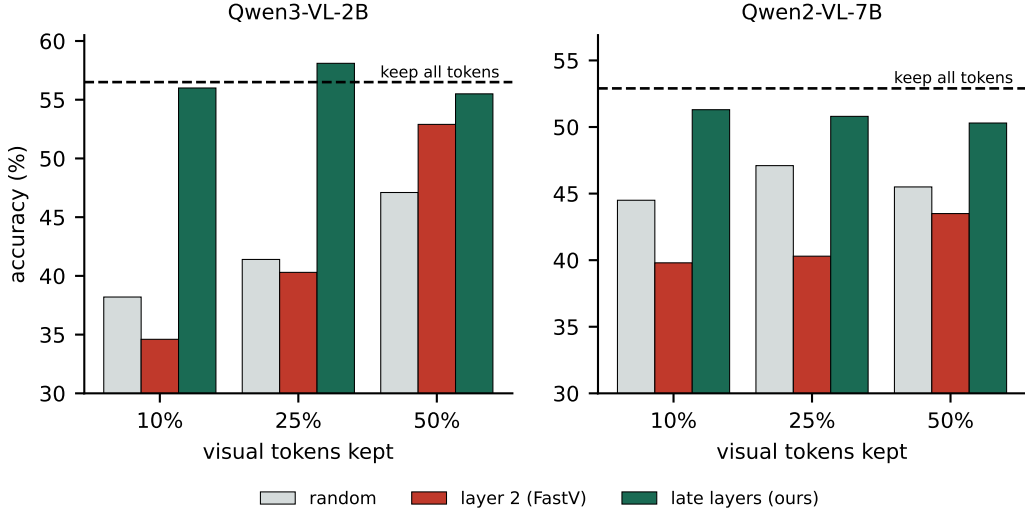


Figure 3: Ranking at layer 2 is worse than ranking at random. Dashed line: keeping every token. A late read-out discards 90% of visual tokens at no measurable cost, on both models.

quality therefore does not predict pruning damage: the claim replicates, its causal story does not, and we state them separately rather than let one carry the other.

6 Mechanism: what cropping actually does

Reading the model’s answer at every layer (Figure 5) separates two explanations. A method that improved reasoning would pull ahead early and widen. Instead the cropped and uncropped arms are indistinguishable through L20, then separate by +15.3 pp at L21 and stay separated. The failing arms jump at L2 — the answer prior forming — and never improve: the uncropped arm’s best score across all 28 layers equals its final answer, so there is no depth at which the answer was available and lost.

The right panel is the causal control. Splitting the same arm by whether its window contained the evidence: +29.7 pp [+19.8, +39.6] where it covers, −7.9 pp [−18.0, +2.2] where it misses — a 37.6 pp swing. The step appears only when the intervention delivers.

Replication. Both the shape and the control hold on Qwen2-VL-7B. Separation is ≈ 0 through L21 and then steps (+8, +16, +13, +12, +12, +10 pp at L22–L27); the index moves by one layer, which we did not require to transfer. The control replicates at +23.8 pp [+13.1, +34.5] where the window covers against +0.0 pp [−11.2, +11.2] where it misses, a 23.8 pp swing. One detail does not generalise: on Qwen3-VL a miss is mildly harmful (−7.9 pp) whereas on Qwen2-VL it is exactly neutral, so the cross-model claim is that the gain appears only where the crop delivers, not that a miss actively hurts. The sharpest sub-claim holds identically on both: the uncropped arm’s maximum over all 28 layers equals its final answer (56.3% and 50.8%), so on neither model is there a depth at which the answer was available and lost.

7 What does not work

Seven internal interventions, each free if it worked, are all null: suppressing the serialisation sink, amplifying attention on the target, residual-stream steering, layer read-out selection, DoLa/DeCo, component ablation, and contrastive decoding steered to the ground-truth region (ceiling +2.6 pp [−0.5, +5.8]). The last is decisive because it aimed perfectly and

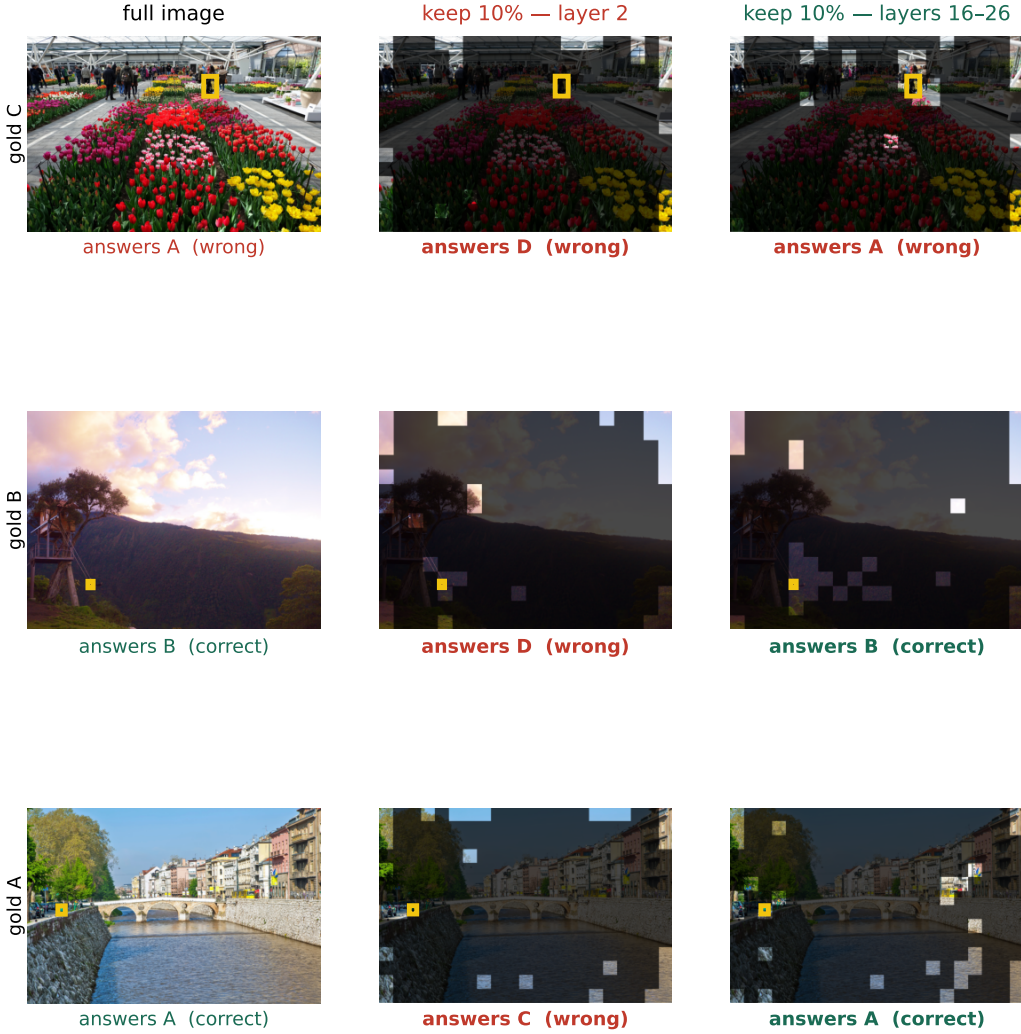


Figure 4: What each ranking actually throws away. Unshaded regions are the 10% of visual tokens kept; everything darkened is discarded before the answer is computed. Yellow box: the queried object. Beneath each panel is the model’s real prediction on that pruned input. Items are the first three in our pool on which the two rankings give different answers — selected on disagreement, never on which ranking turns out to be right, and on the first item both are wrong. Across the full pool of 14, late-layer pruning preserves the full-image answer on 13/14 items and layer-2 pruning on 7/14.

verified the target is causally live — masking the evidence region shifts logits $3.32\times$ more than masking a random region and costs 10.5 pp — and still could not convert it.

The model looks in the right place; the right place is load-bearing; it still cannot answer, because no operation on the residual stream can supply information the encoder never wrote.

Four attempts to regress a continuous geometric quantity from internals also failed (budget from a perfect size oracle; budget from any free signal; window size from target size; attention-space reallocation). Ranking positions is learnable from internals; regressing scale is not.

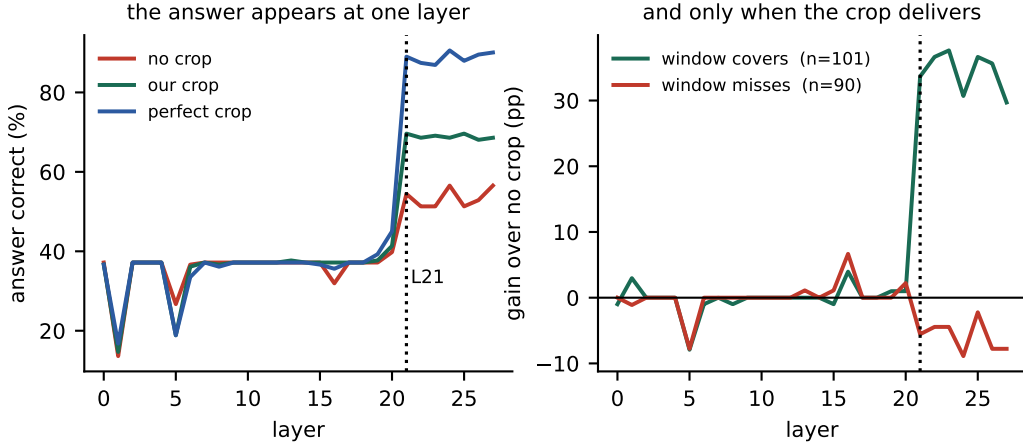


Figure 5: Cropping restores an answer-formation step; it does not improve reasoning. Left: the answer read out at every layer. Right: the same gain split by whether the crop contained the evidence.

8 Limitations

Our crop-placement method improves over an unmodified model on two architectures (+12.0 pp [+4.2, +19.9], +10.5 pp [+2.6, +18.3]) but we have not shown it beats simply spending the same compute on a larger uniform image (+4.7 pp [-3.7, +13.1] and +3.1 pp [-5.2, +11.0]). We report this rather than restricting to the stratum where it clears. Pruning baselines are reimplemented on a common backbone, not run from released checkpoints. Coverage labels require ground-truth boxes and so exist only on V*Bench. Appendix B tags every claim by how many models support it.

Appendix contents

A Experimental protocol • B Replication ledger • C Per-layer structure, four architectures • D The encoding cliff • E The serialisation sink • F The learned read-out: architecture and training • G Prior art at matched budget • H Zero-shot transfer to HR-Bench 4k • I The allocation ledger • J A falsified prediction • K Seven internal interventions • L The vision tower carries no signal • M Further positive results • N Corrections • O Reproducibility

A Experimental protocol

A.1 Benchmarks

V*Bench (Wu & Xie, 2024): 191 four-way multiple-choice questions over high-resolution natural images, in two categories — direct_attributes ($n=115$, one object) and relative_position ($n=76$, a relation between objects). Each item ships a ground-truth bounding box, which is what makes coverage measurable; median target area fraction is 0.0015, i.e. the queried object occupies about one part in seven hundred of the image.

HR-Bench 4k (Wang et al., 2024): 800 rows = 200 instances $\times$ 4 option permutations at 4032², scored by CircularEval (an instance counts only if all four permutations are answered correctly). No boxes ship with it, so no coverage or oracle-capture quantity is computable there and none is quoted. Instances are keyed by image hash as well as question text: 159 unique question strings span 200 instances, and keying on text alone silently pairs 212/800 rows with the wrong image.

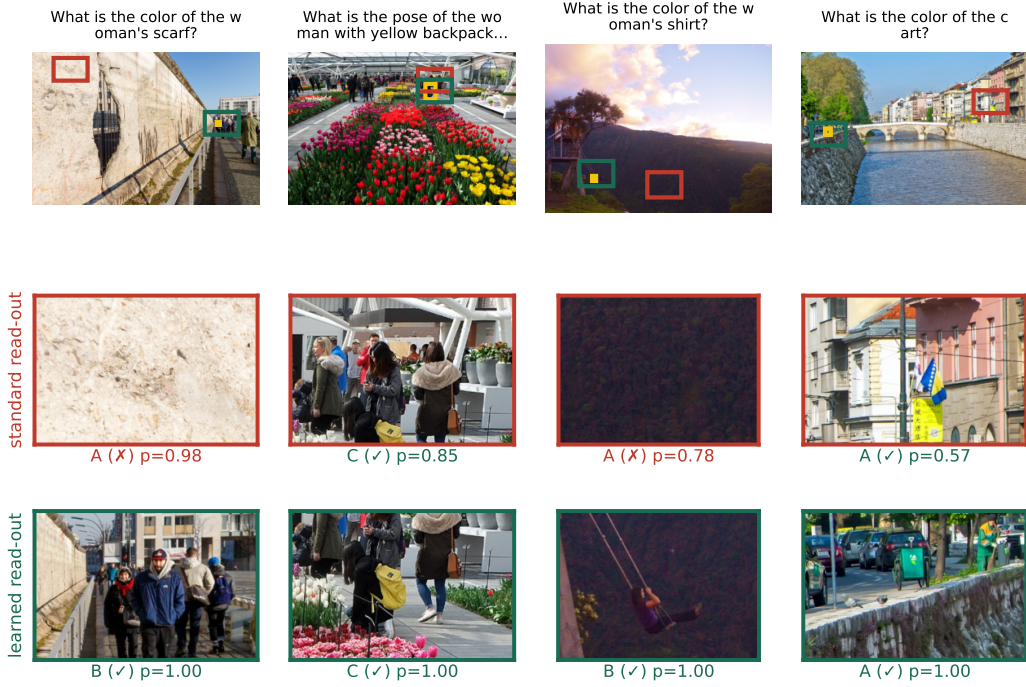


Figure 6: The crops the model is actually fed, and what it actually answers. Top: the source image with the queried object (yellow) and the window each read-out selects. Below: the real cropped input handed to the model, refit to the same 300 tokens, with its prediction and confidence. ✓/× is against the gold answer. Items are the first four in the pool where the standard read-out’s window misses the evidence and the learned one contains it. Note the calibration failure in the leftmost column: the standard read-out crops a featureless stretch of wall and the model answers wrong with $p=0.98$, while the learned crop lands on the people and answers correctly at $p=1.00$. Confidence on a crop that contains nothing is not lower than confidence on a crop that contains the evidence.

Rejected before use. MMBench was considered and dropped: its images are $\sim 512\text{px}$, so there is no allocation headroom, and a null there would have been a benchmark-selection artefact rather than a result.

A.2 Models

Qwen3-VL-2B-Instruct (28 text layers, merge 2), Qwen2-VL-7B-Instruct (28, merge 2), LLaVA-OneVision-7B (28, image_newline separators), LLaVA-NeXT-7B (32, separators). All frozen; no weights are fine-tuned anywhere in this paper.

A.3 Budgets are measured, never computed

A visual-token budget cannot be set analytically: processors merge patches, pad, and round. Every arm is calibrated by iterating on the processor’s own realised grid — rescale, re-measure, repeat — until within 6% of target, and the realised count is logged with every record. Any contrast whose arms differ by more than 10% in median realised tokens is voided rather than reported. In the runs reported here all arms land within 0–6.7%.

A.4 Answer read-out

Four-way accuracy is taken from the option-letter logits at the final position: $p(\text{option}) \propto \exp \text{logsumexp}_j \ell[\text{ids}(\text{option})_j]$, over both the bare and space-prefixed tokenisations of each letter. No generation, no sampling, so results are deterministic given the input.

A.5 Numerical precision

Models load in bfloat16. Loading a bf16 checkpoint in fp16 produces all-NaN logits; because argmax over NaN returns index 0 and the first option is over-represented in the label distribution (71/191 on V*Bench), every arm scores an identical 37.2% with zero-width intervals. This reads as a clean, plausible, uniformly negative result — it cost us a spurious architecture-specificity finding before it was caught. All scripts now assert `torch.isfinite` on logits and attention before writing any record.

A.6 Cross-validation

Every learned component uses grouped k -fold ($k=5$) with folds grouped by item, so an item’s ~ 300 cells never straddle train and test. Layer-selection results are reported out-of-fold; in-sample selection of a “best layer” is the best of 28 candidates on 191 items and is badly optimistic (Appendix N).

A.7 The logit lens

Intermediate layers are read as `lm_head(norm( $h_i$ ))`; the final layer is taken from `model.logits` and never from `norm( $h_{-1}$ )`. HuggingFace has already applied the final norm to the last hidden state, so re-applying it gives reconstruction error 23.47 against 0.06 — corrupting only that layer, which makes the error invisible in layer-wise plots. Both paths are computed and their disagreement asserted, so a zero gap now signals that the correct path is not in use.

A.8 Statistics

All intervals are paired bootstrap over items, 10,000 resamples, percentile method. Where a prediction is stated before a run it is written into the script’s docstring together with its rejection rule, so a result cannot be reinterpreted after the fact. We report intervals rather than p -values and do not correct for multiplicity: the claims that survive are those that replicate on a second architecture, which is a stricter filter than any correction.

B Replication ledger

Our standard is that a claim must hold on more than one model or be rejected rather than caveated. Table 2 is the gate: a claim enters the paper only from the top block.

C Per-layer structure across architectures

Table 3 gives mean `gt_pct` by depth — the rank of the ground-truth cell as a fraction of all cells, where 0.500 is chance by construction and lower is better.

Qwen2-VL’s early and mid layers sit above chance — 15 of its 28 layers are worse than 0.45 — which is why averaging all of them collapses that model to 21.5% (Table 4). Qwen3-VL’s problem is instead its final layer, at 0.529. The general claim is that anti-correlated layers exist and averaging them in is what hurts; the specific identity of the bad layers does not transfer, and we reject the final-layer claim on that basis.

On Qwen3-VL no single layer is stable across folds (in-sample 41.9% collapses to 36.1% out-of-fold, below the block mean it appeared to beat). On Qwen2-VL all five folds inde-

Table 2: Every claim by supporting models.

CLAIM	MODELS	STATUS
Averaging dilutes the read-out	3 of 4, 2 families	survived
Anti-correlated layers exist	4	survived
Layer-2 pruning below random	2	survived
Learned read-out improves proposals	2	survived
Crop method beats unmodified model	2	survived
Answer formation restored by the crop	2	survived
Encoding cliff	2	survived
Serialisation sink	4	survived
“The final layer is anti-correlated”	1 of 4	rejected
“Signed contrast is the mechanism”	1 of 2 tasks	rejected
“No single layer is a good localiser”	false on Qwen2-VL	rejected
Rank-only multi-crop	−3.1 pp vs +5.5 pp predicted	rejected
Crop method beats equal-compute baseline	0 of 2	rejected
Zero-shot transfer to HR-Bench	1	provisional
Evidence region causally live	1	provisional
Seven internal nulls	1	provisional

Table 3: Layer quality by depth. Which layers are anti-correlated differs by architecture; that some are holds on all four.

MODEL	EARLY	MID	LATE	FINAL LAYER
Qwen3-VL-2B	0.456	0.413	0.409	0.529
Qwen2-VL-7B	0.620	0.515	0.392	0.416
LLaVA-OneVision-7B	—	—	—	0.451
LLaVA-NeXT-7B	—	—	—	0.488

pendently choose L21, so that model’s single-layer result is real. This is why we make no general claim that a single layer suffices.

D The encoding cliff

Define tokens-on-target = area fraction $\times$ total visual tokens. This is a sufficient statistic in the sense that arms differing in resolution, crop, and placement fall on one curve when plotted against it.

The same threshold on both, with disjoint confidence intervals on each, across a different model generation, a different vision stack and $3.5\times$ the parameters. On the cleanest subset (localised, single-region, $n=63$) the step is sharper still: 19.0% $\rightarrow$ 78.6%, and 19.0% is below the 25% chance level of a four-way question.

The control carries the claim. The oracle crop is flat across the split on both models at identical values (95.8% below, 100.0% above). Items below the cliff are not harder: given the same 300-token budget spent on the resolved region, both models answer them essentially perfectly. Only the uniform arm steps, and it steps at the same target size in both.

E The serialisation sink

Attention over image positions concentrates on the trailing row boundary. Enrichment is mass share divided by area share, so $1.0\times$ is a fair share; last-token attention, per-

Table 4: Averaging the whole stack versus the deployed block versus a learned reweighting.

MODEL	ALL LAYERS	DEPLOYED BLOCK	LEARNED	BEST SINGLE (out-of-fold)
Qwen3-VL-2B	36.6%	39.3%	45.5%	36.1%
Qwen2-VL-7B	21.5%	35.1%	43.5%	43.5% (L21, all folds)
LLaVA-OneVision-7B	20.4%	25.1%	26.2%	25.1%
LLaVA-NeXT-7B	9.9%	12.6%	18.8%	16.8%

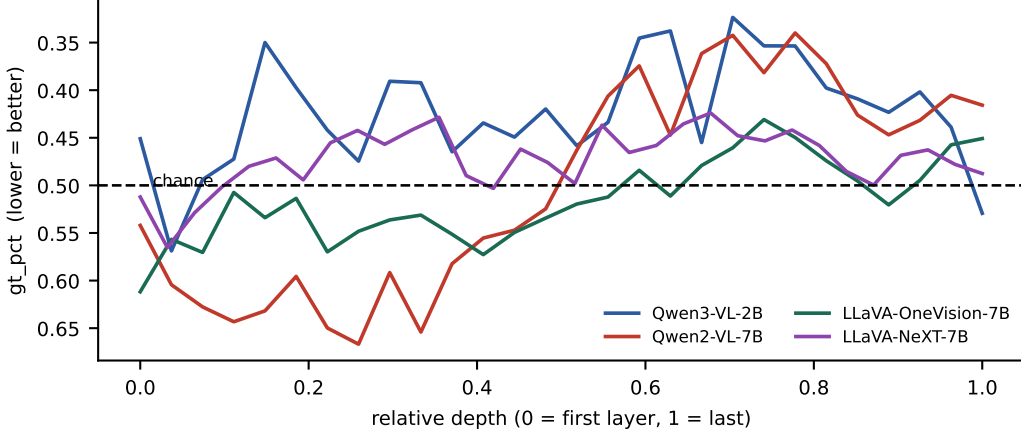


Figure 7: Per-layer localisation quality against relative depth, all four architectures. 0.500 is chance by construction; the axis is inverted so up is better. Every model has layers above chance, but where they sit differs: Qwen2-VL’s first two-thirds are anti-correlated while Qwen3-VL’s problem is its last layer. This is why no fixed layer index generalises and why averaging the whole stack is catastrophic.

layer L_1 -normalised, averaged over a relative block $[0.55L, 0.95L]$ so the measurement is architecture-agnostic. $n=24$ images per model, bootstrap over images.

This is the discriminating measurement between two accounts. A 2-D or corner account predicts nothing special for a newline embedding: it is not a corner, not a pixel, and not part of the image. A serialisation account predicts the sink should sit on the explicit marker wherever one exists. It does, in both LLaVA models, with every CI excluding 1.0. Separator positions are found by matching merged input embeddings against the model’s own `image_newline` parameter, so no unpad logic is reimplemented and nothing depends on our arithmetic.

What we withdrew. An earlier form of this claim added “and rows carry no excess mass.” That half does not survive a five-region split on the LLaVA models, where the bottom row is enriched 2.1–3.0 $\times$. The trailing-boundary half survives; the stronger half is withdrawn.

And what it does not explain. The sink is real and it is not the cause of the proposer’s failure: supplying the learned read-out with explicit last-column and last-row indicators is worth +0.0pp, and removing them costs +0.0pp (Appendix F). Suppressing the sink at inference is likewise null (Appendix K). We report it as a characterised artefact, not as a mechanism.

F The learned read-out: architecture and training

Input. Per visual-token cell: the cell’s L_1 -normalised attention at each of the L layers; its within-layer rank at each layer (so ordering is available as well as magnitude); the deployed

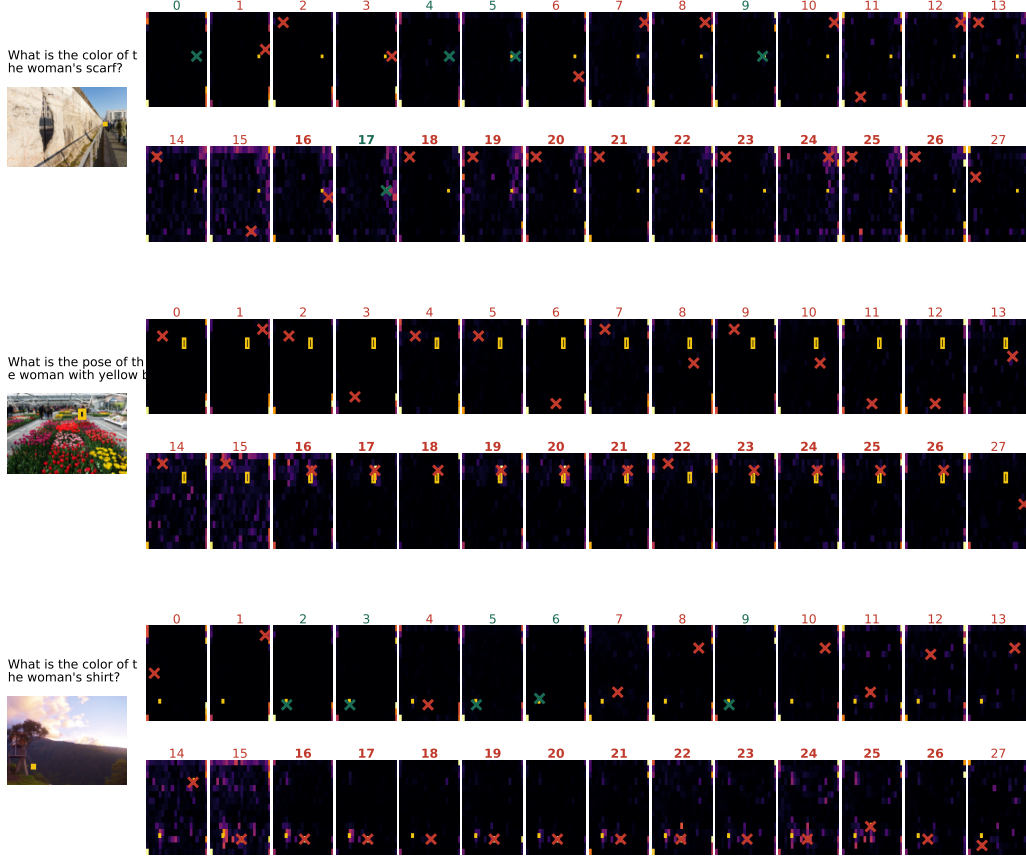


Figure 8: Every layer, three items, one forward pass each. Attention over image positions at each of the 28 layers of Qwen3-VL-2B. Yellow box: the queried object. $\times$: that layer’s ring-masked argmax, green if a $W=0.15$ window there would cover the evidence (the criterion used throughout the paper) and red otherwise. Layers 16–26 — the deployed read-out block — are labelled in bold. All three items are ones where the block-mean read-out fails, and the per-layer view shows why that failure is not a shortage of signal: individual layers do land on the evidence (rows 1 and 3 cover at five layers each), but those layers are L0, 4, 5, 9, 17 and L2, 3, 5, 6, 9 — scattered, and mostly outside the block being averaged. Row 2 covers at no layer, which is the genuinely unrecoverable case. Averaging a fixed contiguous block cannot exploit signal distributed like this, and no fixed block would: the covering layers differ per item.

Table 5: The cliff on single-region items, $n=115$ per model. The oracle arm is the control that makes this a statement about encoding rather than difficulty.

MODEL	CLIFF	UNIFORM		ORACLE CROP	
		below	above	below	above
Qwen3-VL-2B	0.25 tok	32.4% [21, 44]	77.3% [64, 89]	95.8%	100.0%
Qwen2-VL-7B	0.25 tok	36.6% [25, 48]	63.6% [50, 77]	95.8%	100.0%

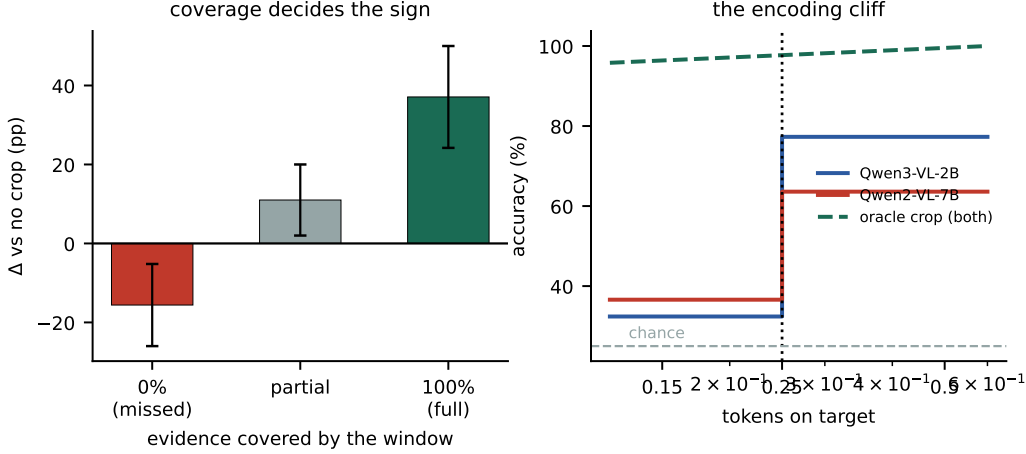


Figure 9: Left: coverage of the evidence set decides the sign of the allocation effect, not merely its size — a window that misses costs -15.6 pp and one that fully covers gains $+37.1$ pp, crossing zero near 25% coverage. Right: the encoding cliff on two architectures, with the oracle crop flat across it. Below the threshold the uniform arm falls below the 25% chance level while the oracle stays near ceiling, which is what makes this a statement about encoding rather than difficulty.

block-mean value and its 3×3 neighbourhood mean; and geometry — row/column fraction, distance to centre, distance to nearest border, and explicit last-column, last-row and first-column indicators. 65 features per cell on a 28-layer model.

Label. The fraction of the ground-truth box falling inside a $W=0.15$ window centred on that cell, with the same edge-clamping the deployed allocator uses, so the label matches what the method would actually crop.

Model. Gradient-boosted trees (HistGradientBoostingRegressor, depth 4, 150 iterations, lr 0.10). Training rows are negative-subsampled — every covering cell plus 30 random non-covering cells per item — while evaluation is always over all cells.

Cost. The head reads attention that pass 1 already computed and runs a tree ensemble over ~ 300 cells on CPU. It adds no forward pass.

Two negative controls. A head given only geometry reaches 3.7% and a head trained on shuffled labels 2.6% — both at the $\sim 2.2\%$ random-cell rate, so the result is not a centre prior and the harness does not leak. A training-free blur of the same map destroys the signal rather than reproducing it ($39.3\% \rightarrow 17.8\%$ at 3×3 , 0.5% at 7×7), so the neighbourhood term is learned local contrast, not smoothing.

The true ceiling. On 88.5% of items some ring-masked cell would have covered the evidence. The head at 52.9% therefore captures 27.6% of the available headroom, not most of it — 35.6pp remains.

Table 6: Trailing-boundary enrichment. Where the row boundary is implicit the sink sits on the last column; where a learned image_newline makes it explicit, the sink sits on the separator token.

MODEL	BOUNDARY	TRAILING	FIRST COL.	INTERIOR
Qwen3-VL-2B	implicit	$3.4 \times [3.0, 3.8]$	$2.0 \times$	$0.7 \times$
Qwen2-VL-7B	implicit	$4.5 \times [4.1, 4.8]$	$0.9 \times$	$0.7 \times$
LLaVA-OneVision-7B	image_newline	$2.0\text{--}2.3 \times$	—	—
LLaVA-NeXT-7B	image_newline	$2.0\text{--}2.3 \times$	—	—

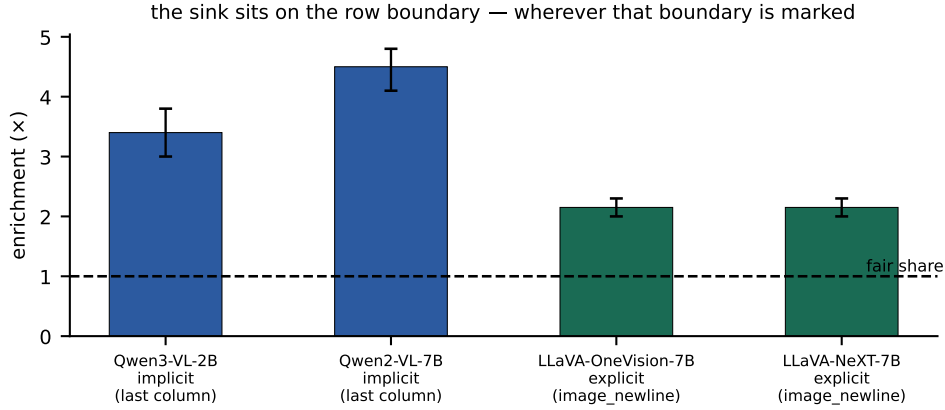


Figure 10: Trailing-boundary enrichment on four architectures. Where the row boundary is implicit the sink sits on the last column; where a learned image_newline makes it explicit, it sits on the separator token itself. Every interval excludes a fair share.

G Prior art at matched budget

The comparison this literature does not run: charge every policy for every image token across every pass, and score it against what the same realised budget buys as one uniform image, measured in-run on the same items.

Zoom Eye has the highest raw accuracy in the table and still does not beat one uniform image of its own budget. At 4K the same comparison is significant: -15.3 pp $[-18.6, -11.8]$ at $n=800$.

Scope. These are reimplementations on a common backbone, not released checkpoints; V*/SEAL’s search module is absent. On V*Bench no individual margin is significant at $n=191$. The defensible statement is that no policy significantly beats the budget axis at its own spend, which includes ours.

H Zero-shot transfer to HR-Bench 4k

The learned read-out trained on V*Bench, applied to HR-Bench 4k with nothing refitted — not the window, not the layer block, not the ring mask, not B_0 . HR-Bench ships no boxes, so the head cannot have been fitted there even in principle.

The component transfers: $+4.9$ pp $[+1.8, +8.1]$ per-row and $+6.5$ pp on CircularEval over the standard read-out, rising to $+8.5$ pp $[+4.0, +13.0]$ on single-region questions, across a different benchmark at $2.7 \times$ the resolution with zero refitting. The allocator does not: -12.4 pp against the compute-matched bar. At 4K, spending the budget beats allocating it, and a better proposal narrows that gap without closing it.

Table 7: Feature-group ablation on Qwen3-VL-2B. The depth profile carries the result; the sink indicators contribute nothing.

FEATURE SET	COVERAGE	Δ
deployed block mean (incumbent)	39.3%	—
deployed map only (value + 3×3 + position)	45.0%	+5.8 pp
+ sink indicators	44.5%	+5.2 pp
+ depth profile	51.8%	+12.6 pp
+ within-layer ranks (full head)	52.9%	+13.6 pp
full head minus sink indicators	52.9%	+13.6 pp

Table 8: V*Bench, $n=191$, Qwen3-VL-2B. “Bar” is the uniform sweep interpolated at that arm’s own realised token spend.

POLICY	PASSES	TOKENS	ACC	BAR	MARGIN
uniform@300	1.00	296	56.5%	—	—
uniform@600	1.00	598	63.9%	—	—
uniform@1200	1.00	1182	71.7%	—	—
grounding crop	2.00	595	57.6%	63.8%	−6.2 pp
Zoom Eye (tree search)	9.00	2664	75.9%	77.8%	−1.9 pp
ours, fixed policy	2.00	591	61.3%	63.8%	−2.5 pp
ours, gated	1.40	414	62.8%	60.0%	+2.8 pp
oracle (uses GT box)	1.00	301	92.7%	56.7%	+35.9 pp
random placement	1.00	296	33.5%	56.6%	−23.0 pp

Internal control. On the 160 rows where the two read-outs chose the same cell the arms are the same computation and split +0.0 pp [+0.0, +0.0]; on the 640 where they differ, +6.1 pp [+2.2, +10.0]. The effect lives only where the proposals diverge.

I The allocation ledger: what we could not make work

Four attempts to regress a continuous geometric quantity from model internals, all failed:

- Token budget from target size. With a perfect ground-truth area oracle, the rule $b \geq t^*/\text{area}$ loses to flat uniform in 11 of 12 cells, each scored against the uniform ladder at that controller’s own mean spend. The oracle controller’s required spend is nearly flat in target size (3212/2146/2318/1615 tokens across a $20\times$ range of area fraction), which is why size cannot drive it.
- Token budget from any free pass-1 signal. 17 features (attention concentration, entropy, spread, peak ratios; logit-lens confidence, margin, cross-layer agreement, settling depth; image geometry). Out-of-fold $\rho = 0.254$, but exact-rung accuracy is 9.4% against a 35.6% majority baseline — worse than a constant, with mean error 1.98 rungs where a rung is a doubling. The required-budget distribution is bimodal (68/191 items need the smallest budget, 37 need more than the ladder offers) and no free signal makes that call.
- Window size from target size. Across four quartiles spanning a $100\times$ range of target area, the best window is 0.15 in every quartile. A per-item oracle window is worth +11.0 pp after a stability check, so the headroom is real — but no predictor we tried reaches it.
- Attention-space reallocation. Amplifying attention on the ground-truth cells recovers 16% of what cropping the same region buys (+5.8 pp against the crop’s +36.2 pp). Every practical arm is indistinguishable from a random control. The oracle amplification set is one cell of 294: the target is sub-token, so there is nothing to amplify.

Table 9: HR-Bench 4k, $n=800$ rows / 200 instances, balanced 400 single / 400 cross.

ARM	PER-ROW	CIRCULAREVAL
uniform@300	52.8%	37.5%
uniform@600 (compute-matched bar)	59.9%	46.0%
standard read-out crop	42.6%	26.5%
learned read-out crop	47.5%	33.0%
random placement	38.9%	20.0%

Ranking positions is learnable from internals. Regressing scale is not. Every method in this paper that works is a ranking method, and every one that failed asked the model how much rather than where.

A ceiling that was noise. Per-item-best over a budget ladder reads 90.1% against the plateau’s 84.8%, which looks like +5.2pp of headroom. It is not: 22.5% of items flip correct→wrong going up the ladder, so “ever correct” harvests noise on a four-way question. Requiring monotone stability collapses the ceiling to exactly the plateau. We apply the same test to the window-size ceiling above, which survives it.

J Rank-only multi-crop: a falsified prediction

The ledger above suggests a method that only ranks: take the top- k separated cells, split the budget evenly, show all k in one pass — no size or budget decision anywhere. We priced it before running: union coverage of the top-4 and above the encoding cliff at 75 tokens each is 63.4% against single-crop’s 52.9%, i.e. +10.5 pp of answerable items at identical total budget, turning over at $k=5$. At the measured coverage exchange rate this predicts +5.5 pp.

Measured: -3.1 pp $[-9.4, +3.1]$. The prediction had the wrong sign, so the model behind it is wrong, not merely imprecise. The exchange rate was estimated on single crops, where coverage decides whether the evidence is visible at all; with four crops, three are distractors and the account has no term for distractor cost. Ranking still works inside the format — our top-4 beats four random crops by +27.7 pp — but the format costs more than the extra coverage buys.

This contradicted an earlier in-project result of +7.9 pp for multi-crop, so before recording the negative we tested the clearest difference between the two setups: the connector text between images. Re-running with the descriptive connector gave identical accuracy on all three arms while 0/25 probability vectors matched byte-for-byte — the prompt changes the output without flipping decisions. The deficit is real, not a prompting artefact.

K Seven internal interventions

Each would have been free if it worked; each is null.

1. Serialisation-sink suppression. Biasing attention away from the trailing row boundary is indistinguishable from biasing away random or interior image tokens (-0.5 pp), while the same operation on prompt tokens costs -17.3 pp — so the machinery bites, and image-token attention is simply not scarce in the direction of removal.
2. Attention amplification on the target. Ground-truth-targeted amplification gains +5.8 pp $[+1, +10]$ at its best bias strength, 16% of what cropping the same region buys. Every practical arm (top-1, top-5, top-15, deployed window) is indistinguishable from a random control, and at the strongest bias all arms collapse by 9–14 pp including the control.
3. Residual-stream steering. A shared direction exists (cosine 0.53–0.75) and an item’s own displacement is 47% reachable at L20, but the transferable vector gives +0.0 pp.

4. Layer read-out selection for the answer. No layer under uniform encoding exceeds 52.9% against the oracle crop’s 92.6%.
5. DoLa / DeCo. Null against a correctly-computed baseline. Against the double-normalised baseline of Appendix N they appeared favourable, which is how that bug was found.
6. Component ablation. Null.
7. Contrastive decoding steered to the ground-truth region. The decisive one, because it aims perfectly. Masking the evidence region shifts logits $3.32\times$ more than masking a random region and costs 10.5 pp of accuracy — the region is unambiguously load-bearing — and yet contrast on it reaches a ceiling of +2.6 pp $[-0.5, +5.8]$, and -5.5 pp against its own compute-matched bar. The gain is larger where the window covers (+3.0 pp) than where it misses (+1.1 pp), so the effect is evidence-linked; it is simply far too small to matter.

The seventh retires the family. The previous six could each be dismissed as steering to the wrong place; this one steered to the ground truth, verified the place was causally live, and still could not convert it.

L The vision tower carries no localisation signal

If the vision encoder’s own attention localised the target, resolution could be allocated at encode time for two vision encodes and one language-model forward, instead of the two forwards that crop-and-re-encode pays. We tested it. Qwen3-VL-2B’s tower is 24 blocks, 16 heads, patch 16, merge 2; per-patch salience is attention received, averaged over heads and queries, pooled to the merged grid the language model sees.

Table 10: Vision-tower localisation, $n=191$, same ring mask and coverage definition as the language-model read-out.

PROPOSER	COVERAGE
random ring-masked cell (chance, 200 draws/item)	2.3%
best single vision layer	1.6%
learned signed combination over 24 vision layers (OOF)	1.0%
mean of all 24 vision layers	0.5%
language-model read-out (deployed)	39.3%
language-model read-out (learned)	45.5%

Every vision-tower arm is at or below chance. Not weak — absent. The vision encoder never sees the question, so there is no question-independent salience that suffices when the target is small and the question selects among several plausible objects.

Two consequences. Single-pass foveation steered by vision salience is dead at step one. And the learned reweighting reproduces the same weight shape here (12 positive / 12 negative, final layer -0.382) while buying nothing, which shows that depth contrast amplifies signal that is present; it does not create it.

M Positive results not required by the main argument

These are measured, survive their own controls, and are reported for completeness even though the paper’s argument does not depend on them.

M.1 The exchange rate that motivates allocation at all

At an identical 300-token budget, placing the tokens on the evidence reaches 92.7% against uniform’s 56.5%, and no uniform rung catches it even at 7957 tokens (84.8%). The exchange

rate is therefore at least $26\times$ in tokens, and it is a lower bound set by the model’s own ceiling rather than by our sweep. This is what makes where worth studying; the rest of the paper is about why that potential is so hard to realise.

M.2 Coverage is the mediator, and it decides the sign

Stratifying by how much of the evidence set the reallocated window covers:

Table 11: Allocation effect by coverage, V*Bench. Difficulty is held fixed within stratum.

COVERAGE	n	NO CROP	CROP	Δ
0% (missed entirely)	96	52.1%	36.5%	-15.6 pp [-26.0, -5.2]
100% (full)	62	58.1%	95.2%	+37.1 pp [+24.2, +50.0]

The sign flips near 25% coverage, and half of all windows miss. Within the wrong-at-uniform stratum the gap reaches 70.8 pp: covered items score 83.8% and missed items 13.0%, below chance. It replicates on Qwen2-VL (-9.4/+41.8 pp). This is the quantity Section 6’s causal control is built on.

M.3 Where the method pays: below the cliff, not above it

Splitting the learned read-out’s gain by tokens-on-target:

Table 12: The method is a fix for sub-token evidence, and the cliff predicts its own operating regime.

STRATUM	n	LEARNED	STANDARD	Δ
below the cliff (< 0.15)	58	67.2%	53.4%	+13.8 pp [+1.7, +25.9]
cliff zone ($0.15\text{--}0.25$)	25	64.0%	48.0%	+16.0 pp [+4.0, +32.0]
above the cliff (≥ 0.25)	108	70.4%	66.7%	+3.7 pp [-3.7, +11.1]

Above the cliff the target is already resolved and placement barely matters; below it the crop is the only operation that resolves the target. This also refutes a tempting explanation for the weaker 4K transfer — “below the cliff there is nothing to re-rank” is false by its own sign.

M.4 A free reliability signal

The peak value of the ring-masked block-mean map — a number the read-out already computes on its way to an argmax — predicts whether the proposal will cover the evidence at AUROC 0.788 (any overlap) and 0.839 (at least half), at no additional cost. A two-pass confidence signal reaches 0.835 but requires cropping first, which is the expense. Gating on the free signal improves a competitor as well as our own arm: Zoom Eye recovers +5.7 pp at 4K on 53% fewer tokens.

M.5 Multiple windows in a single pass

Packing several separated windows into one forward pass at the same total token budget beats a single window by **+7.9 pp** [+1.0, +14.7] on V*Bench at 601 vs 590 tokens, and replicates at +8.0 pp on HR-Bench 4k — $2.7\times$ the image scale. This is not the same as Appendix J’s failure: there the budget was split k ways at a fixed window size chosen for single-crop use, which pushed covered targets back under the cliff. The distinction is that multiple windows help when the per-window budget stays above the encoding threshold and hurt when it does not, which is the cliff making a quantitative prediction about a method.

M.6 Read-out choices validated rather than assumed

The deployed last-token read-out is the best of seven candidate query positions; attention from the noun that names the target is near chance (gt_pct 0.367), so localisation here is not lexical lookup. The relative layer block transfers: on Qwen2-VL, 8/8 of the individually-best layers fall inside the block transferred from Qwen3-VL, with gt_pct 0.041 inside versus 0.390 outside.

M.7 The sink is not a register token

Two controls rule out the obvious alternative. Transposing the image moves the enrichment with the serialisation order rather than with image content, and an interior-replay control places 1.1% of mass where chance predicts 3.3% — i.e. below chance. The sink is present at layer 0, before any language-model processing.

N Corrections made during this work

We list these because several changed conclusions, and because the mechanism that caught each one is the part worth reusing.

- A double-normalised logit lens (Appendix A.7) manufactured a “+5.1 pp free read-out gain”, an evidence-dependent late-layer bias, a letter-concentration finding, and a favourable comparison against DoLa. Caught because a second code path computed the same quantity and disagreed — 100% agreement at L24, 82.2% at the final layer.
- A transfer result called negative from a prefix. HR-Bench is ordered by category; the first 244 rows were 65% relational, the stratum where our method does nothing. The balanced $n=800$ result reversed the sign ($+0.0 \rightarrow +4.9$ pp, CI clear of zero).
- Two mechanisms proposed for a failure that did not exist. Both refuted by our own data — “the map is flat at 4K” (peak concentration 0.0228 vs 0.0269, only $1.18\times$ apart) and “below the cliff there is nothing to re-rank” (the gain is largest below the cliff).
- A capture fraction against a restricted ceiling. Quoted as 65% of an oracle-among-top-5 gap; against the true ceiling (does any cell cover: 88.5%) it is 27.6%. The loose version also implied the method was near its limit when 35.6 pp of headroom remained.
- An AUROC computed on an unmasked map (0.576); ring-masked as everywhere else in the paper it is 0.745. This briefly made a second signal look competitive when it is substantially worse.
- In-sample layer selection read 41.9% and collapsed to 36.1% out-of-fold — below the block mean it appeared to beat, inverting the conclusion for that model.
- A pre-registered mechanism prediction that failed (Section 5): the layer-2 pruning penalty should have been larger on the model whose early layers rank the target worse. It is smaller. The claim replicates; the causal story does not, and we report them separately.
- A significance test computed on mismatched items inflated a pooled margin from $+3.9$ pp $[-0.3, +8.2]$ to a spurious “+5.2 pp $[+1.1, +9.4]$, significant” by taking the bar from all items while scoring an easier subset.
- A row-segmentation error treated LLaVA’s separator-free base image as “row 0”, placing 59.8% of positions in a bucket capped at $\sim 5\%$ and manufacturing a spurious $3.0\times$ result.
- An objective that was maximised by doing nothing: selecting a gate threshold by “accuracy per pass” drives the gate to never fire (0.565/1.00 beats 0.670/2.00). Replaced by a target firing rate with the whole cost–accuracy curve reported.

O Reproducibility

Every figure in this paper is generated from measured data by a script in the release; no number is typed into a figure by hand. Figures 1, 4 and 6 are rendered from a dedicated inference pass that stores the attention maps, the exact pruning masks, the crop pixels handed to the model, and the model’s real predictions, so each panel is an artefact of a forward pass rather than an illustration. Each such figure states its item-selection rule in its caption, and the pruning figure selects on disagreement between rankings rather than on which ranking turns out to be correct.

An executable suite of 86 invariants encodes every retraction and every corrected number above. Each is a named test asserting the shape of a result — that a control sits at chance, that a CI includes zero, that an in-sample figure exceeds its out-of-fold counterpart — so a claim the data stopped supporting fails a test rather than surviving in prose. The suite runs in seconds, with no GPU, model or network access.

References

- Liang Chen, Haozhe Zhao, Tianyu Liu, Shuai Bai, Junyang Lin, Chang Zhou, and Baobao Chang. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In European Conference on Computer Vision (ECCV), 2024.
- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. DoLa: Decoding by contrasting layers improves factuality in large language models. In International Conference on Learning Representations (ICLR), 2024.
- Sarthak Jain and Byron C. Wallace. Attention is not explanation. In NAACL-HLT, 2019.
- Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- Wenbin Wang, Liang Ding, Minyan Zeng, Xiabin Zhou, Li Shen, Yong Luo, and Dacheng Tao. Divide, conquer and combine: A training-free framework for high-resolution image perception in multimodal large language models. arXiv preprint arXiv:2408.15556, 2024.
- Penghao Wu and Saining Xie. V*: Guided visual search as a core mechanism in multimodal LLMs. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In International Conference on Learning Representations (ICLR), 2024.